

● HARD

● GEOMETRY AND TRIGONOMETRY

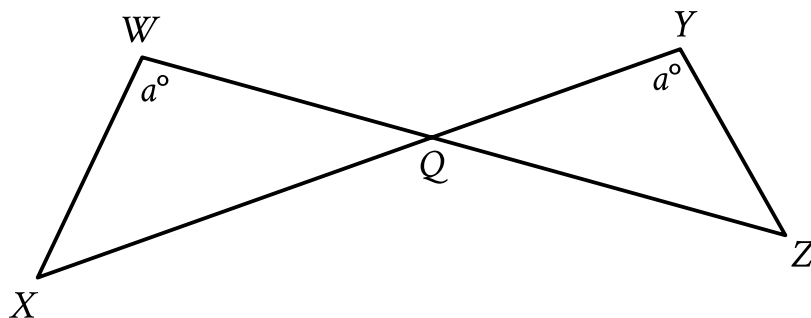
Geometry and Trigonometry

03

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- Triangle WXY has a common vertex with triangle XZY .
- The common vertex is Q .
- For triangle WXY :
 - The measure of angle W is a° .
- For triangle XZY :
 - The measure of angle Y is a° .
- A note indicates the figure is not drawn to scale.

In the figure shown, WZ and XY intersect at point Q , $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of YZ ?

- A. 18
- B. 36
- C. 120
- D. 200

Q2

GRID-IN

GEOMETRY AND TRIGONOMETRY

A rectangular poster has an area of 360 square inches. A copy of the poster is made in which the length and width of the original poster are each increased by 20%. What is the area of the copy, in square inches?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN

GEOMETRY AND TRIGONOMETRY

A circle in the xy -plane has a diameter with endpoints $2, 4$ and $2, 14$. An equation of this circle is $x - 2^2 + y - 9^2 = r^2$, where r is a positive constant. What is the value of r ?

STUDENT-PRODUCED RESPONSE

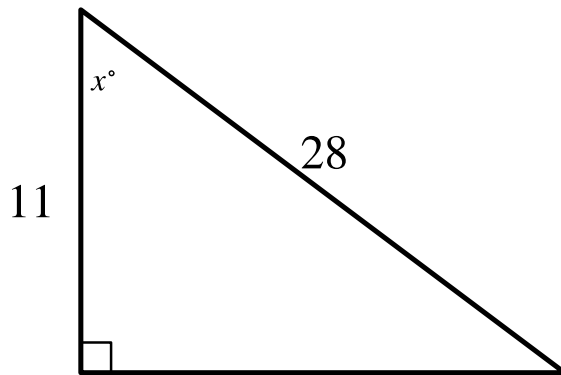
.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The length of the hypotenuse is 28.
- The length of the leg adjacent to the angle with measure x° is 11.
- A note indicates the figure is not drawn to scale.

In the triangle shown, what is the value of $\cos x^\circ$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GEOMETRY AND TRIGONOMETRY

Triangle XYZ is similar to triangle RST such that X , Y , and Z correspond to R , S , and T , respectively. The measure of $\angle Z$ is 20° and $2XY = RS$. What is the measure of $\angle T$?

- A. 2°
- B. 10°
- C. 20°
- D. 40°

Q6

GEOMETRY AND TRIGONOMETRY

Circle A has a radius of $3n$ and circle B has a radius of $129n$, where n is a positive constant. The area of circle B is how many times the area of circle A ?

- A. 43
- B. 86
- C. 129
- D. 1,849

Q7

GEOMETRY AND TRIGONOMETRY

Point F lies on a unit circle in the xy -plane and has coordinates $1, 0$. Point G is the center of the circle and has coordinates $0, 0$. Point H also lies on the circle and has coordinates $-1, y$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

- A. $\frac{27\pi}{2}$
- B. $\frac{29\pi}{2}$
- C. 24π
- D. 25π

Q8

GEOMETRY AND TRIGONOMETRY

For two acute angles, $\angle Q$ and $\angle R$, $\cos Q = \sin R$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$, respectively. What is the value of x ?

- A. 5
- B. 19
- C. 23
- D. 29

Q9

GRID-IN

GEOMETRY AND TRIGONOMETRY

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $9x - 560^\circ$. The sum of the measures of one of the acute angles and three of the obtuse angles is $-18x + w^\circ$. What is the value of w ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The floor of a ballroom has an area of 600 square meters. An architect creates a scale model of the floor of the ballroom, where the length of each side of the model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of the ballroom. What is the area, in square meters, of the scale model?

- A. 6
- B. 10
- C. 60
- D. 150